CI31/CY31

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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS – APRIL 2023

Program	:	**B.E :- Common to CSE (CY) / CSE (AI & ML)**	**Semester**	:	**III**
Course Name	:	**Linear Algebra , Laplace Transforms and Optimization**	**Max. Marks**	:	**100**
Course Code	:	**CI31 / CY31**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:
- Answer one full question from each unit.

UNIT - I

1. a) Evaluate (i) $L\left\{e^t \int_0^t \frac{\sin t}{t} dt\right\}$ (ii) $L\left\{(t^5+1)^2\right\}$. CO1 (06)

b) Using convolution theorem find the inverse Laplace transform of CO1 (06)

$$\frac{s}{(s+2)(s^2+9)}$$

c) Solve the differential equation $\frac{d^2y}{dt^2}+4\frac{dy}{dt}+4y=e^{-t}$ given $y(0)=y'(0)=0$ by the method of Laplace transform. CO1 (08)

2. a) Given $L\left\{2Sin\sqrt{t}\right\} == \frac{\sqrt{\pi}}{2s^{3/2}}e^{-1/4s}$,show that $L\left\{\frac{\cos\sqrt{t}}{\sqrt{t}}\right\} == \frac{\sqrt{\pi}}{s^{1/2}}e^{-1/4s}$. CO1 (06)

b) Find the inverse Laplace transform of $\frac{4s+5}{(s+1)^2(s+2)}$ by the method of partial fraction. CO1 (06)

c) The coordinates (x,y) of a particle moving along a plane curve at any point t, are given by $y'+2x=\sin 2t$; $x'-2y=\cos 2t$; $t>0$. If at $t=0,\ x=1$ and $y=0$, then show by using Laplace Transforms that the particle moves along the curve $4x^2+4xy+5y^2=4$. CO1 (08)

UNIT - II

3. a) A manufacturer produces nuts and bolts for industrial machinery. It takes 1 hour of work on machine A and 3 hours on machine B to produce a package of nuts, while it takes 3 hours on machines A and 1 hour's on machine B to produce a package of bolts. He earns a profit of Rs.2.50 per package on nuts and Rs.1 per package on bolts. Form a linear programming problem to maximize his profit, if he operates each machine for at most 12 hours a day. CO2 (05)

b) Find the maximum value of $Z=3x_1+4x_2$ subject to the constraints $4x_1+2x_2\le 80$, $4x_1+2x_2\le 80$, $2x_1+5x_2\le 180$, $x_1,x_2\ge 0$, by solving LPP using graphical method. CO2 (05)

c) Using Big – M method, solve the LPP to minimize $z=2x_1+x_2$, subject to $3x_1+x_2=3$, $4x_1+3x_2\ge 6$, $x_1+2x_2\le 3$, $x_1\ge 0$, $x_2\ge 0$. CO2 (10)

4. a) Construct the dual of the LPP to maximize $z = 4x_1 + 9x_2 + 2x_3$, subject to $2x_1 + 3x_2 + 2x_3 \le 7$, $3x_1 - 2x_2 + 4x_3 = 5$, $x_1 \ge 0$, $x_2 \ge 0$, $x_3 \ge 0$. CO2 (05)

b) i) Explain 2-phase Simplex method in LPP. CO2 (05)

ii) Explain the procedure to construct the dual problem in LPP.

c) Using simplex method, solve the LPP to maximize $z = 5x_1 + 3x_2$ subject to the constraints $x_1 + x_2 \le 2$, $5x_1 + 2x_2 \le 10$, $3x_1 + 8x_2 \le 12$, $x_1 \ge 0$, $x_2 \ge 0$. CO2 (10)

UNIT - III

5. a) Define composition of two matrix transformations. Determine the matrix that defines a reflection in the y-axis, followed by rotation through an angle $\pi/6$ and then by the dilation of factor 3/2 . Find the image of the point $\begin{bmatrix} 3 \\ 6 \end{bmatrix}$ under this transformation. CO3 (06)

b) Show that a linear operator $T : R^2 \to R^2$ defined by $T(X) = AX$ rotates a vector X through an angle θ about the origin if $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. CO3 (07)

c) Define linear transformation. Determine whether the transformation T(x, y, z) = (x + 2y, x + y + z, 3z) is linear or not. CO3 (07)

6. a) Show that the set of vector {(1, 5, 2), (-1, -1, 2), (1, 4, 1)} spans R^3. Express the vector (1, 3, -2) in terms of the spanning set. CO3 (06)

b) Determine the kernel and range of the transformation T(x, y) = (4x, x-y, 3y) of $R^2 \to R^3$ and hence verify Rank nullity theorem. CO3 (07)

c) Consider the bases B = {(1, 2), (3, -1)} and B' = {(3, 1), (5, 2)} of R^2. Find the transition matrix from B to B'. If u is a vector such that $u_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, find $u_{B'}$. CO3 (07)

UNIT- IV

7. a) Let $u_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}, u_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}, y = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $w = span\{u_1, u_2\}$. Show that $\{u_1, u_2\}$ is an orthogonal basis. Compute $\mathrm{Proj}_W y$. CO4 (06)

b) Find the complete solution of $A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$. CO4 (07)

c) Find the least square solution to $Ax = b$, where $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$. CO4 (07)

8. a) Find the QR factorization of $A=\begin{bmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$. CO4 (10)

b) Find the dimension and basis for four fundamental subspaces of $A=\begin{bmatrix}-2 & 4 & -2 & -4\\ 2 & -6 & -3 & 1\\ -3 & 8 & 2 & -3\end{bmatrix}$.Hence verify the orthogonality of the each subspaces. CO4 (10)

UNIT - V

9. a) Find the steady state solution of the Markov matrix $A=\begin{bmatrix}0.95 & 0.03\\ 0.05 & 0.97\end{bmatrix}$ CO5 (03)

b) Determine Unitary Matrix *P* that diagonalize the given matrix: CO5 (07)

$$A=\begin{bmatrix}0 & 2+i\\ 2-i & 4\end{bmatrix}$$

c) Find an Singular value decomposition of the matrix $A=\begin{bmatrix}1 & 1\\ 3 & -3\end{bmatrix}$. CO5 (10)

10. a) Determine whether the matrix $A=\begin{bmatrix}3 & 7-4i & -2+5i\\ 7+4i & -2 & 3+i\\ -2-5i & 3-i & 4\end{bmatrix}$ is Hermitian. CO5 (03)

b) Orthogonally diagonalize the matrix $A=\begin{bmatrix}16 & -4\\ -4 & 1\end{bmatrix}$. CO5 (07)

c) Find the principal components of the matrix $A=\begin{bmatrix}4 & 8 & 13 & 7\\ 11 & 4 & 5 & 14\end{bmatrix}$. CO5 (10)

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